

Backpropagation in a CNN, Part 2: Flatten, Max Pooling and Convolution

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1. Overview

Key point: Going backwards, flatten is undone by a reshape, max pooling sends each gradient to the position that held the maximum, ReLU lets gradients through only where its input was positive, and the convolution gives $\partial L/\partial b_1$ as the sum of $\partial L/\partial Z_1$ and $\partial L/\partial W_1 = X * \partial L/\partial Z_1$: the filter's gradient is itself a convolution.

The [part 1 Note](#) set up a small CNN, wrote its forward equations and found the gradients of its output node. To train the filter we still need to go back through flatten, max pooling, ReLU and the convolution. This Note finds each of these backward steps.

$$\partial L/\partial W_1 = \text{input } X \text{ convolved with } \partial L/\partial Z_1$$

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input X , 3×3		$\partial L/\partial Z_1$, 2×2		$\partial L/\partial W_1$, 2×2																	

top-left entry: lay $\partial L/\partial Z_1$ on the highlighted window of X , multiply cell by cell, add:

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z_{11}}x_{11} + \frac{\partial L}{\partial z_{12}}x_{12} + \frac{\partial L}{\partial z_{21}}x_{21} + \frac{\partial L}{\partial z_{22}}x_{22}$$

Figure 1: The filter's gradient is a convolution: slide $\partial L/\partial Z_1$ over the input X exactly as a filter would slide

Figure 1 shows the final result. The Notebook checks every formula against TensorFlow's `GradientTape`, then trains the small CNN on real MNIST digits using only these formulas.

2. Prerequisites

- The [part 1 Note](#): the network, its forward equations, the chains of derivatives, and $\partial L/\partial Z_2 = a_2 - y$.
- The [convolution operation Note](#) and the [pooling Note](#).
- The [memoization Note](#): backpropagation stores one gradient per step and reuses it.

3. The plan

Key point: Six derivatives remain. The first two factors, $\partial L / \partial A_2 \cdot \partial A_2 / \partial Z_2 = a_2 - y$, are already known from part 1.

Recall the network of the part 1 Note: $X (6 \times 6) \rightarrow$ convolution with $W_1 (3 \times 3)$ and $b_1 \rightarrow Z_1 (4 \times 4) \rightarrow \text{ReLU} \rightarrow A_1 (4 \times 4) \rightarrow$ max pooling $\rightarrow P_1 (2 \times 2) \rightarrow$ flatten $\rightarrow F (4 \times 1) \rightarrow Z_2 = W_2 F + b_2 \rightarrow A_2 = \sigma(Z_2) \rightarrow$ loss L .

The chains for the filter's weights and bias are

$$\frac{\partial L}{\partial W_1} = \underbrace{\frac{\partial L}{\partial A_2} \frac{\partial A_2}{\partial Z_2}}_{a_2 - y} \cdot \frac{\partial Z_2}{\partial F} \cdot \frac{\partial F}{\partial P_1} \cdot \frac{\partial P_1}{\partial A_1} \cdot \frac{\partial A_1}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial W_1}$$

and the same with $\partial Z_1 / \partial b_1$ at the end. Part 1 found the first two factors. We now take the rest one at a time, from the output backwards.

The gradient flowing back has a name at every step: $\partial L / \partial Z_2$, then $\partial L / \partial F$, $\partial L / \partial P_1 (2 \times 2)$, $\partial L / \partial A_1 (4 \times 4)$, $\partial L / \partial Z_1 (4 \times 4)$, and finally $\partial L / \partial W_1$ and $\partial L / \partial b_1$. Each has the shape of its tensor. As in the [memoization Note](#), each one is computed once and reused for the next step.

4. Back through the output node: $\partial Z_2 / \partial F = W_2$

Key point: $\partial L / \partial F = W_2^\top (a_2 - y)$, a 4×1 column, the shape of F .

From $Z_2 = W_2 F + b_2 = w_1 f_1 + w_2 f_2 + w_3 f_3 + w_4 f_4 + b_2$, the derivative with respect to each input f_k is its weight w_k . So $\partial Z_2 / \partial F$ is made of the 4 weights of W_2 .

The gradient with respect to F must have the shape of F , 4×1 , because there is one gradient for every value of F . W_2 is 1×4 , so we use its transpose:

$$\frac{\partial L}{\partial F} = \underbrace{W_2^\top}_{4 \times 1} \underbrace{(a_2 - y)}_{1 \times 1}$$

5. Back through flatten: reshape

Key point: Flatten only rearranges numbers, so going back means rearranging the gradients the same way: reshape the 4×1 gradient into 2×2 , the shape of P_1 .

Flatten has no trainable parameters, and it does no arithmetic: it takes the 2×2 values of P_1 and lays them out as a 4×1 column F . Each value of F is one value of P_1 , moved. So there is nothing to differentiate in the usual sense; the backward step simply reverses the forward one.

1. **In words:** put each gradient back where its value came from.

2. **Formula:**

$$\frac{\partial L}{\partial P_1} = \text{reshape}\left(\frac{\partial L}{\partial F}, \text{shape of } P_1\right)$$

3. **Example:** if $\partial L/\partial F = (0.1, -0.2, 0.3, 0.4)^\top$, then

$$\frac{\partial L}{\partial P_1} = \begin{bmatrix} 0.1 & -0.2 \\ 0.3 & 0.4 \end{bmatrix}$$

So far: $\partial L/\partial P_1 = \text{reshape}(W_2^\top(a_2 - y))$, four numbers.

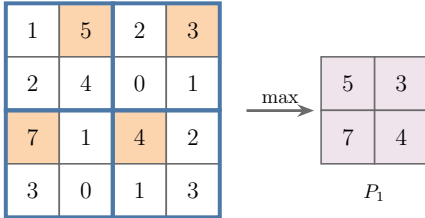
6. Back through max pooling: route to the maximum

Key point: Only the maximum of each window reached the output, so only it receives a gradient. Every other position of A_1 gets 0.

Max pooling has no trainable parameters either. Going backwards we must turn the 2×2 gradient $\partial L/\partial P_1$ into the 4×4 gradient $\partial L/\partial A_1$: the reverse of max pooling.

In every window, max pooling passed one value forward, the maximum, and dropped the other three. The dropped values had no role in the prediction $\hat{y}$, so no role in the loss: their gradient is 0. A small change in the maximum, on the other hand, changes P_1 by exactly the same amount, so the maximum receives the window's gradient unchanged.

forward: A_1 and its maxima



backward: route each gradient to its maximum

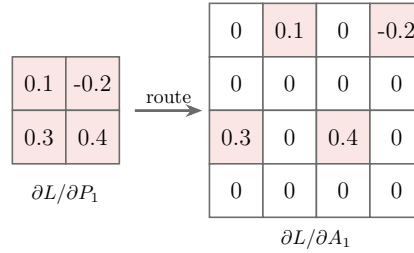


Figure 2: Left: max pooling of A_1 ; the maximum of each 2×2 window is shaded. Right: going backwards, each of the four gradients of $\partial L/\partial P_1$ is placed where its window's maximum was; all other positions get 0

1. **In words:** ask A_1 where the maximum of each window was, put the window's gradient there, and 0 everywhere else.

2. **Formula:** for position (m, n) of A_1 , inside the window that produced $P_{1,xy}$:

$$\frac{\partial L}{\partial A_{1,mn}} = \begin{cases} \frac{\partial L}{\partial P_{1,xy}} & \text{if } A_{1,mn} \text{ is the maximum of its window} \\ 0 & \text{otherwise} \end{cases}$$

3. **Example (Figure 2):** the maxima of A_1 are 5, 3, 7 and 4. With $\partial L/\partial P_1 = \begin{bmatrix} 0.1 & -0.2 \\ 0.3 & 0.4 \end{bmatrix}$:

$$\frac{\partial L}{\partial A_1} = \begin{bmatrix} 0 & 0.1 & 0 & -0.2 \\ 0 & 0 & 0 & 0 \\ 0.3 & 0 & 0.4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The indices x, y number the positions of P_1 and m, n those of A_1 ; they help when writing the code. To do this backward step, the forward pass must remember where each maximum was (CS231n notes call these positions the “switches”). When a window holds two equal maxima, as in an all-zero window after ReLU, the Notebook sends the gradient to the first of them.

7. Back through ReLU: a mask

Key point: ReLU’s derivative is 1 where $Z_1 > 0$ and 0 elsewhere, so $\partial L/\partial Z_1 = \partial L/\partial A_1$ wherever Z_1 was positive, and 0 elsewhere.

$A_1 = \text{ReLU}(Z_1)$ applies $\max(0, z)$ to each of the 16 cells of Z_1 separately. Its derivative is 1 for a positive input and 0 for a negative one (see the [activation functions Note](#), section 8):

$$\frac{\partial A_{1,xy}}{\partial Z_{1,xy}} = \begin{cases} 1 & \text{if } Z_{1,xy} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Multiplying cell by cell:

$$\frac{\partial L}{\partial Z_1} = \frac{\partial L}{\partial A_1} \odot \mathbb{K}[Z_1 > 0]$$

where $\odot$ means cell-by-cell multiplication. We now have $\partial L/\partial Z_1$, a 4×4 matrix: how the loss changes with each value of the feature map.

8. Back through the convolution

Key point: Every value of Z_1 contains b_1 once, so $\partial L/\partial b_1$ is the sum of all of $\partial L/\partial Z_1$. Each weight w_{ij} multiplies a window of X , so $\partial L/\partial W_1$ is X convolved with $\partial L/\partial Z_1$.

The convolution layer, unlike flatten and max pooling, has trainable parameters, the filter W_1 and its bias b_1 . To see the pattern clearly we shrink the example: a 3×3 input and a 2×2 filter, so Z_1 and $\partial L/\partial Z_1$ are 2×2 . Everything else stays the same.

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}, \quad W_1 = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

The forward convolution gives four equations:

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{21}w_{21} + x_{22}w_{22} + b_1$$

$$z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{22}w_{21} + x_{23}w_{22} + b_1$$

$$z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{31}w_{21} + x_{32}w_{22} + b_1$$

$$z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{32}w_{21} + x_{33}w_{22} + b_1$$

8.1 The bias: a sum

Key point: $\partial L/\partial b_1 = \sum \partial L/\partial Z_1$, a single number like b_1 .

b_1 appears in all four values of Z_1 , so a change in b_1 reaches the loss along four paths, and the chain rule adds them:

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_{11}} \frac{\partial z_{11}}{\partial b_1} + \frac{\partial L}{\partial z_{12}} \frac{\partial z_{12}}{\partial b_1} + \frac{\partial L}{\partial z_{21}} \frac{\partial z_{21}}{\partial b_1} + \frac{\partial L}{\partial z_{22}} \frac{\partial z_{22}}{\partial b_1}$$

Each $\partial z/\partial b_1$ is 1, because b_1 enters every equation with coefficient 1. So:

1. **In words:** add up all the values of $\partial L/\partial Z_1$.
2. **Formula:**

$$\frac{\partial L}{\partial b_1} = \sum_{x,y} \frac{\partial L}{\partial Z_{1,xy}}$$

3. **Example:** with $\partial L/\partial Z_1 = \begin{bmatrix} 0.5 & -1 \\ 0.25 & 2 \end{bmatrix}$, $\partial L/\partial b_1 = 0.5 - 1 + 0.25 + 2 = 1.75$. TensorFlow gives 1.75 (Notebook).

8.2 The filter: a convolution

Key point: $\partial L/\partial w_{ij}$ adds up $\partial L/\partial z$ times the input pixel that w_{ij} multiplied in each equation. The pattern is exactly the convolution of X with $\partial L/\partial Z_1$.

Each weight also appears in all four equations, each time multiplying a different pixel. For w_{11} these pixels are $x_{11}, x_{12}, x_{21}, x_{22}$, so

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z_{11}} x_{11} + \frac{\partial L}{\partial z_{12}} x_{12} + \frac{\partial L}{\partial z_{21}} x_{21} + \frac{\partial L}{\partial z_{22}} x_{22}$$

and the same reasoning for the other weights gives

$$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial z_{11}}x_{12} + \frac{\partial L}{\partial z_{12}}x_{13} + \frac{\partial L}{\partial z_{21}}x_{22} + \frac{\partial L}{\partial z_{22}}x_{23}$$

$$\frac{\partial L}{\partial w_{21}} = \frac{\partial L}{\partial z_{11}}x_{21} + \frac{\partial L}{\partial z_{12}}x_{22} + \frac{\partial L}{\partial z_{21}}x_{31} + \frac{\partial L}{\partial z_{22}}x_{32}$$

$$\frac{\partial L}{\partial w_{22}} = \frac{\partial L}{\partial z_{11}}x_{22} + \frac{\partial L}{\partial z_{12}}x_{23} + \frac{\partial L}{\partial z_{21}}x_{32} + \frac{\partial L}{\partial z_{22}}x_{33}$$

These look complex, but there is a pattern. In $\partial L/\partial w_{11}$, the 2×2 matrix $\partial L/\partial Z_1$ is laid on the top-left 2×2 window of X , multiplied cell by cell and summed. In $\partial L/\partial w_{12}$ the same matrix is laid one step to the right; in $\partial L/\partial w_{21}$ one step down; in $\partial L/\partial w_{22}$ down and right. This pattern is the convolution operation, with $\partial L/\partial Z_1$ playing the role of the filter (Figure 1).

1. **In words:** convolve the input with the gradient of the feature map.
2. **Formula:**

$$\frac{\partial L}{\partial W_1} = X * \frac{\partial L}{\partial Z_1}$$

3. **Example:** $X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $\partial L/\partial Z_1 = \begin{bmatrix} 0.5 & -1 \\ 0.25 & 2 \end{bmatrix}$. The top-left entry is

$$0.5(1) - 1(2) + 0.25(4) + 2(5) = 9.5$$

and the full result is $\begin{bmatrix} 9.5 & 11.25 \\ 14.75 & 16.5 \end{bmatrix}$, exactly what `GradientTape` returns (Notebook).

The shapes agree: a 3×3 input convolved with a 2×2 matrix gives $3 - 2 + 1 = 2$, the shape of W_1 . In the 6×6 network of part 1, X is 6×6 and $\partial L/\partial Z_1$ is 4×4 , which gives $6 - 4 + 1 = 3$: the 3×3 shape of the filter.

Extra: Goodfellow et al. (2016, §9.5, equation 9.11) give the same result for any number of channels and any stride: the gradient for kernel entry $K_{i,j,k,l}$ is a sum over output positions of the incoming gradient times the input value that entry multiplied. For a network with more convolution layers, the gradient must also pass back to the layer's input, $\partial L/\partial X$; that backward pass “is also a convolution (but with spatially-flipped filters)” (CS231n notes).

9. The whole backward pass

Key point: Five steps: $a_2 - y \rightarrow$ multiply by $W_2^T \rightarrow$ reshape $\rightarrow$ route through max pooling $\rightarrow$ mask with ReLU $\rightarrow$ sum (bias) and convolve with X (filter).

Putting the steps together, for one image:

Step	Gradient	Shape	Rule
Output node	$\partial L / \partial Z_2 = a_2 - y$	1×1	log loss and sigmoid (part 1)
Into F	$\partial L / \partial F = W_2^\top (a_2 - y)$	4×1	weights of the output node
Flatten	$\partial L / \partial P_1 = \text{reshape}(\partial L / \partial F)$	2×2	reverse the reshape
Max pooling	$\partial L / \partial A_1$	4×4	route to the maximum, 0 elsewhere
ReLU	$\partial L / \partial Z_1 = \partial L / \partial A_1 \odot \mathbb{I}[Z_1 > 0]$	4×4	mask
Convolution	$\partial L / \partial b_1 = \sum \partial L / \partial Z_1$	1×1	sum
Convolution	$\partial L / \partial W_1 = X * \partial L / \partial Z_1$	3×3	convolution

With a batch of images, each image goes through these steps and the gradients are averaged over the batch, as for W_2 in part 1.

9.1 Checking against TensorFlow

Key point: On a real 28×28 MNIST digit, all four gradients match `GradientTape` to within 3×10^{-16} .

The Notebook builds the same kind of network at full MNIST size: a 28×28 digit, one 3×3 filter (26×26 feature map), ReLU, 2×2 max pooling (13×13), flatten (169 values), one sigmoid node. The task: is the digit a 1 ($y = 1$) or a 7 ($y = 0$)? The hand-coded backward pass and TensorFlow's automatic gradients agree for W_1 , b_1 , W_2 and b_2 , with the largest difference below 3×10^{-16} , rounding error. Of the 676 values of $\partial L / \partial A_1$, 169 are non-zero: one per pooling window.

9.2 Training with these formulas only

Key point: Mini-batch gradient descent with the hand-derived gradients takes test accuracy from 54.7% to 99.4% in 2 epochs.

Using nothing but the formulas of this Note and part 1, the Notebook trains the network with mini-batch gradient descent: batches of 32 images, learning rate 0.1, 2 epochs over 13,007 training images of 1s and 7s.

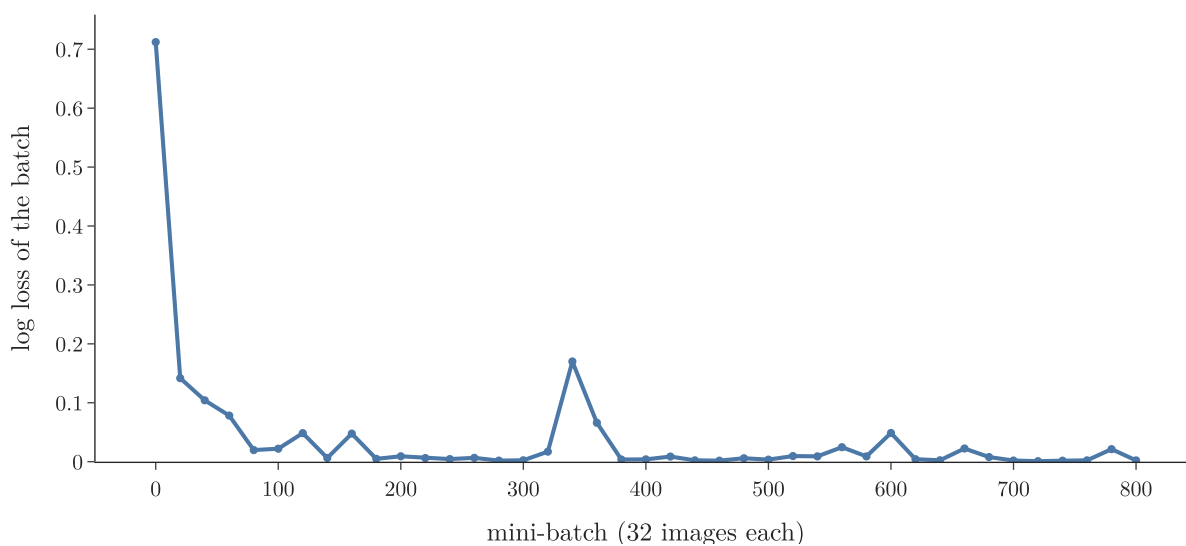


Figure 3: The loss of the batch while training with the hand-derived gradients; it drops from 0.71 to near 0 within the first few hundred mini-batches

The test accuracy on 2,163 unseen 1s and 7s goes from 54.7% before training to 99.0% after the first epoch and 99.4% after the second. Backpropagation through convolution, max pooling and flatten works exactly as derived.

10. Summary

Layer	Trainable parameters	Backward rule
Dense (output node)	W_2, b_2	$\partial L / \partial W_2 = (a_2 - y) F^T$; pass $W_2^T(a_2 - y)$ back
Flatten	none	reshape to the input's shape
Max pooling	none	gradient to each window's maximum, 0 elsewhere
ReLU	none	multiply by 1 where the input was positive, 0 elsewhere
Convolution	W_1, b_1	$\partial L / \partial b_1 = \sum \partial L / \partial Z_1$; $\partial L / \partial W_1 = X * \partial L / \partial Z_1$

- Layers without parameters still pass the gradient back; they just have nothing to update.
- Flatten: reshape. Max pooling: route to the maximum. ReLU: mask.
- The bias of a filter collects the sum of the gradient over the whole feature map.
- The gradient of a filter is the convolution of the input with the gradient of the feature map.
- Checked against `GradientTape`, and enough to train a CNN from scratch.

11. Sources

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. \$9.5, equations 9.11–9.13 (gradients of a convolution with respect to the kernel and the input).
- Stanford CS231n course notes, “Convolutional Neural Networks”, sections on the backward pass of convolution and of pooling, cs231n.github.io/convolutional-networks.
- TensorFlow API documentation: `tf.GradientTape`, [tensorflow.org/api_docs/python/tf/GradientTape](https://www.tensorflow.org/api_docs/python/tf/GradientTape)

12. Key terms

Term	Meaning
Backward pass	Computing the gradient of the loss for every tensor, from the output back to the input
Gradient routing	In max pooling's backward pass, sending each gradient only to the position of its window's maximum
Switches	The stored positions of the maxima, needed for max pooling's backward pass
ReLU mask	The 0/1 matrix $\mathbb{I}[Z > 0]$ that ReLU's backward pass multiplies by

Term	Meaning
$\partial L / \partial b_1$	The sum of the gradient over the whole feature map
$\partial L / \partial W_1$	The convolution of the input with the gradient of the feature map